

Maintain AI

Formal Technology Names & Architectural Definitions

Public technical reference — formal names, acronyms, short descriptions and full definitions

Overview

This document presents the formal terminology used to describe the architectural constructs developed as part of Maintain AI. It is designed as a standalone technical reference for researchers, engineers, reviewers, collaborators and other readers.

1. Formal Name Reference Chart

No.	Acronym	Formal name	Short definition
1	MHICPIA	Maintain AI Heterogeneous Industrial Cyber-Physical Intelligence Architecture	Unified system architecture connecting machines, gateways, telemetry, intelligence, maintenance, safety and workforce operations.
2	HMIRM	Heterogeneous Machine Intelligence Representation Model	Structured representation of different machines, capabilities, telemetry, state, health and maintenance properties.
3	HITSNA	Heterogeneous Industrial Telemetry Semantic Normalization Architecture	Translates machine-specific measurements and representations into consistent platform telemetry semantics.
4	AIITIA	Authenticated Idempotent Industrial Telemetry Ingestion Architecture	Authenticated, persistent and retry-safe industrial telemetry ingestion.
5	OIRITIA	Organization-Isolated Real-Time Industrial Telemetry Architecture	Organization-scoped and authorization-controlled realtime telemetry distribution.
6	MAAOIA	Machine-Assignment-Aware Operational Intelligence Architecture	Operational authorization based on users, roles and assigned machines.
7	HMT-IIA	Hierarchical Multi-Tenant Industrial Intelligence Isolation Architecture	Hierarchical isolation of organizations, users, machines, telemetry and intelligence.
8	PIIEA	Persistent Industrial Intelligence Evidence Architecture	Persistent lifecycle for predictions, conditions, alerts and maintenance evidence.
9	MAI-RA	Model-Agnostic Industrial Intelligence Runtime Architecture	Stable runtime interface supporting interchangeable intelligence models.
10	DLE-IIA	Distributed Local-Edge Industrial Intelligence Inference Architecture	Local/edge inference coordinated with cloud intelligence services.
11	BAAI-AIM	Bounded Assistive Artificial Intelligence Architecture for Industrial Maintenance	Assistive AI with explicit authority boundaries.
12	PCASIA	Predictive Condition-Aware Safety Intervention Architecture	Condition/prediction evidence connected to a defined intervention pathway.
13	CPMIFA	Closed-Loop Predictive Maintenance Intelligence Feedback Architecture	Prediction-to-maintenance-to-outcome intelligence workflow.
14	OC-PMLA	Outcome-Conditioned Predictive Maintenance Learning Architecture	Maintenance outcomes linked to preceding predictive conditions.
15	SITVA	Synthetic Industrial Telemetry Validation Architecture	Reproducible synthetic telemetry and fault-scenario validation.

2. Full Technical Definitions

1. MHICPIA

Maintain AI Heterogeneous Industrial Cyber-Physical Intelligence Architecture

Short definition: Unified system architecture connecting machines, gateways, telemetry, intelligence, maintenance, safety and workforce operations.

Formal definition: The umbrella architecture defines the separation and interaction of the physical-machine, acquisition, gateway, platform, intelligence, safety and workforce layers as one industrial intelligence system.

2. HMIRM

Heterogeneous Machine Intelligence Representation Model

Short definition: Structured representation of different machines, capabilities, telemetry, state, health and maintenance properties.

Formal definition: HMIRM provides a common machine abstraction while preserving machine-specific characteristics, allowing different machine classes to participate in common platform services and intelligence workflows.

3. HITSNA

Heterogeneous Industrial Telemetry Semantic Normalization Architecture

Short definition: Translates machine-specific measurements and representations into consistent platform telemetry semantics.

Formal definition: HITSNA separates device-specific acquisition from platform-level meaning by defining how different names, units, fields and machine measurements are represented consistently for downstream services.

4. AIITIA

Authenticated Idempotent Industrial Telemetry Ingestion Architecture

Short definition: Authenticated, persistent and retry-safe industrial telemetry ingestion.

Formal definition: AIITIA defines the ingestion path in which telemetry is associated with its machine and organizational context, authenticated, identified, persisted and handled safely when events are retried or delivered more than once.

5. OIRITIA

Organization-Isolated Real-Time Industrial Telemetry Architecture

Short definition: Organization-scoped and authorization-controlled realtime telemetry distribution.

Formal definition: OIRITIA combines live telemetry delivery with server-side organizational isolation so that realtime machine information is distributed only within the authorized operational context.

6. MAAOIA

Machine-Assignment-Aware Operational Intelligence Architecture

Short definition: Operational authorization based on users, roles and assigned machines.

Formal definition: MAAOIA makes machine assignment part of the operational authorization model, so access to industrial resources is determined by the relationship between identity, organization, role and machine assignment.

7. HMT-IIA

Hierarchical Multi-Tenant Industrial Intelligence Isolation Architecture

Short definition: Hierarchical isolation of organizations, users, machines, telemetry and intelligence.

Formal definition: HMT-IIA defines the relationships through which industrial resources remain scoped to their authorized organization and operational hierarchy rather than being treated as globally accessible platform objects.

8. PIIEA

Persistent Industrial Intelligence Evidence Architecture

Short definition: Persistent lifecycle for predictions, conditions, alerts and maintenance evidence.

Formal definition: PIIEA turns transient intelligence events into durable records that can be connected to subsequent alerts, maintenance activities, technician actions and outcomes.

9. MAI-RA

Model-Agnostic Industrial Intelligence Runtime Architecture

Short definition: Stable runtime interface supporting interchangeable intelligence models.

Formal definition: MAI-RA separates platform consumers from individual model implementations, allowing different forecasting, anomaly-detection or predictive models to be loaded and executed through a common intelligence boundary.

10. DLE-IIA

Distributed Local-Edge Industrial Intelligence Inference Architecture

Short definition: Local/edge inference coordinated with cloud intelligence services.

Formal definition: DLE-IIA defines how selected intelligence workloads can execute close to machines or gateways while cloud services continue to provide broader coordination, persistence, management and workforce access.

11. BAAI-AIM

Bounded Assistive Artificial Intelligence Architecture for Industrial Maintenance

Short definition: Assistive AI with explicit authority boundaries.

Formal definition: BAAI-AIM separates AI assistance from authoritative system responsibilities, keeping telemetry authority, authorization, system state and safety mechanisms under designated system components rather than an AI assistant.

12. PCASIA

Predictive Condition-Aware Safety Intervention Architecture

Short definition: Condition/prediction evidence connected to a defined intervention pathway.

Formal definition: PCASIA provides an architectural pathway from machine conditions or predictive evidence toward safety intervention while separating that pathway from ordinary informational predictions and interface behavior.

13. CPMIFA

Closed-Loop Predictive Maintenance Intelligence Feedback Architecture

Short definition: Prediction-to-maintenance-to-outcome intelligence workflow.

Formal definition: CPMIFA closes the operational loop by connecting observations and predictions with alerts, maintenance actions, technician activity, observed outcomes and subsequent evidence.

14. OC-PMLA

Outcome-Conditioned Predictive Maintenance Learning Architecture

Short definition: Maintenance outcomes linked to preceding predictive conditions.

Formal definition: OC-PMLA structures the relationship between predictive conditions and observed maintenance resolutions so that outcomes can become evidence for model evaluation, error analysis and future learning workflows.

15. SITVA

Synthetic Industrial Telemetry Validation Architecture

Short definition: Reproducible synthetic telemetry and fault-scenario validation.

Formal definition: SITVA enables controlled industrial telemetry and fault scenarios to pass through the operational pipeline so ingestion, intelligence, alerting and maintenance workflows can be tested repeatedly without requiring every scenario to occur physically.

3. Architectural Relationship

The constructs are designed as a connected architecture rather than isolated features. HMIRM defines the machine representation; HITSNA provides common telemetry semantics; AIITIA provides the ingestion boundary; OIRITIA, MAAOIA and HMT-IIA govern access and isolation; MAI-RA and DLE-IIA define intelligence execution boundaries; PIIEA, CPMIFA and OC-PMLA connect intelligence to maintenance evidence and outcomes; PCASIA defines a safety-intervention pathway; BAAI-AIM defines the authority boundary for assistive AI; and SITVA provides reproducible validation.

At the system level, these constructs are unified by MHICPIA, the overall Maintain AI heterogeneous industrial cyber-physical intelligence architecture.

4. Terminology Note

These formal names identify the specific architectural constructs and implementations developed within Maintain AI. The names themselves should not be interpreted as claims that Maintain AI invented the underlying general technologies or research fields. Any academic claim of novelty should be supported by implementation evidence, prior-art comparison and experimental validation.